Genome-wide fusion-hotspot cohort scan: msk_impact_50k_2026

544 of 3919 genes passed the >= 5-patient recurrence gate; 17 FDR-significant (q<0.05).

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Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
TMPRSS2	auto	ok	1088	33.33	6.243	0.0006243	yes	0.587
EGFR	auto	ok	466	70	74.55	0.0006243	yes	0.3311
BRAF	curated	ok	247	92.64	91.57	0.0006243	yes	0.5048
EML4	auto	ok	224	97.51	53.36	0.0006243	yes	0.7578
RET	curated	ok	219	85.38	92.27	0.0006243	yes	0.5767
ETV6	curated	ok	213	81.01	77.27	0.0006243	yes	0.5736
FGFR3	auto	ok	194	97.37	94.04	0.0006243	yes	0.5665
FGFR2	curated	ok	188	93.1	85.82	0.0006243	yes	0.4161
KIF5B	auto	ok	95	97.7	96.88	0.0006243	yes	0.7389
CD74	auto	ok	74	68.42	89.39	0.0006243	yes	0.6589
KIAA1549	auto	ok	44	97.56	50	0.0006243	yes	0.6214
PRKACA	auto	ok	35	96	96.67	0.0006243	yes	0.7742
ATF1	auto	ok	30	84	90	0.001092	yes	0.4549
TPM3	auto	ok	14	100	64.29	0.001092	yes	0.5971
ROS1	curated	ok	204	72.73	90.68	0.001942	yes	0.4662
NAB2	auto	ok	111	71.43	69.01	0.009199	yes	0.4295
ALK	curated	ok	321	97.41	96.31	0.01665	yes	0.6585
NTRK3	curated	ok	71	98.21	96.49	0.07592	no	0.509
ERG	curated	ok	863	32.96	97.68	0.1007	no	0.5331
NCOA3	auto	ok	33	50	75	0.1294	no	0.1846
STAT6	auto	ok	60	100	6	0.3011	no	0.3991
NTRK1	curated	ok	113	72.73	84.21	0.3201	no	0.3559
RELA	auto	ok	11	66.67	100	0.4225	no	0.4962
RBM10	auto	ok	36	75	71.43	0.4366	no	0.3861
AGAP3	auto	ok	6	100	100	0.4498	no	0.6166
RAD51B	auto	ok	16	50	50	0.4726	no	0.3288
CDKN2B	auto	ok	40	66.67	20	0.5249	no	0.2992
POLD1	auto	ok	36	33.33	46.15	0.7353	no	0.2623
EWSR1	auto	ok	415	78.35	11.3	1	no	0.3727
TP53	auto	ok	258	30.77	27.91	1	no	0.2279

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
NF1	auto	ok	247	27.78	47.73	1	no	0.1374
ARID1A	auto	ok	148	44.44	37.93	1	no	0.08826
SMARCA4	auto	ok	143	36.84	41.67	1	no	0.1849
TACC3	auto	ok	138	97.06	48.48	1	no	0.5671
KMT2C	auto	ok	134	14.29	53.57	1	no	0.112
KMT2D	auto	ok	128	25	48.57	1	no	0.04628
CREBBP	auto	ok	121	50	32.61	1	no	0.1314
FLI1	curated	ok	121	68.22	94.02	1	no	0.5989
WT1	auto	ok	119	93.94	4.808	1	no	0.5423
NOTCH1	auto	ok	115	57.89	60.87	1	no	0.2025
NOTCH3	auto	ok	101	25	59.09	1	no	0.09913
CDK12	auto	ok	99	50	39.02	1	no	0.2874
EP300	auto	ok	98	14.29	31.82	1	no	0.07645
STK11	auto	ok	97	14.29	30.43	1	no	0.1403
NOTCH2	auto	ok	92	25	53.85	1	no	0.2176
ERBB2	auto	ok	88	35	43.18	1	no	0.1084
TSC2	auto	ok	88	42.86	25	1	no	0.2475
BRD4	auto	ok	86	11.11	39.29	1	no	0.1141
PTEN	auto	ok	86	50	25	1	no	0.1487
DOT1L	auto	ok	85	55.56	36.36	1	no	0.09687
PBRM1	auto	ok	85	16.67	50	1	no	0.158
DNMT1	auto	ok	80	45.45	20	1	no	0.1047
BRCA2	auto	ok	79	14.29	63.64	1	no	0.07525
MET	auto	ok	73	89.47	75	1	no	0.1674
POLE	auto	ok	73	36.36	31.82	1	no	0.1119
ATM	auto	ok	72	12.5	32	1	no	0.08901
CTNNB1	auto	ok	72	60	44.44	1	no	0.0742
KDM6A	auto	ok	71	50	100	1	no	0.1127
KMT2B	auto	ok	68	66.67	46.67	1	no	0.07198
BAP1	auto	ok	67	42.86	33.33	1	no	0.1776
BRCA1	auto	ok	66	25	52	1	no	0.06825
KMT2A	auto	ok	65	40	62.5	1	no	0.06373
MGA	auto	ok	65	33.33	46.67	1	no	0.09197
ATRX	auto	ok	64	50	42.86	1	no	0.1717
FGFR1	auto	ok	64	58.33	50	1	no	0.127
CDH1	auto	ok	58	50	85.71	1	no	0.1426
PTPRD	auto	ok	57	25	42.86	1	no	0.07841

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
RTEL1	auto	ok	57	50	37.5	1	no	0.1395
PIK3R1	auto	ok	55	50	33.33	1	no	0.1399
DNAJB1	auto	ok	54	100	2.564	1	no	0.3699
RECQL4	auto	ok	54	50	41.94	1	no	0.07871
CCDC6	auto	ok	53	94.23	1.887	1	no	0.426
MDC1	auto	ok	52	50	57.14	1	no	0.1016
GLI1	auto	ok	51	57.14	60	1	no	0.11
STAT3	auto	ok	51	100	11.11	1	no	0.08312
AGO2	auto	ok	49	46.15	31.82	1	no	0.2518
FANCA	auto	ok	49	30	35	1	no	0.1045
MTOR	auto	ok	49	33.33	30	1	no	0.08587
RNF43	auto	ok	48	60	38.89	1	no	0.2447
ELF3	auto	ok	47	50	55.56	1	no	0.1141
NF2	auto	ok	46	20	11.11	1	no	0.1661
ERBB3	auto	ok	45	50	58.82	1	no	0.2257
NSD1	auto	ok	45	50	36.36	1	no	0.1118
TFE3	auto	ok	45	87.5	80.56	1	no	0.2642
KDM5A	auto	ok	44	50	55.56	1	no	0.1494
PTPRS	auto	ok	44	25	38.46	1	no	0.0784
PAX8	auto	ok	43	42.86	80	1	no	0.276
PIK3C2G	auto	ok	43	50	33.33	1	no	0.1048
RPTOR	auto	ok	43	37.5	18.18	1	no	0.1579
ERBB4	auto	ok	42	100	40	1	no	0.09893
MAP3K1	auto	ok	42	33.33	30	1	no	0.1227
CARM1	auto	ok	39	40	27.78	1	no	0.2132
EZH2	auto	ok	39	40	7.143	1	no	0.2622
FOXP1	auto	ok	39	50	80	1	no	0.2137
INPPL1	auto	ok	39	57.14	53.85	1	no	0.1421
RAD21	auto	ok	38	100	55.56	1	no	0.3195
AXL	auto	ok	37	75	25	1	no	0.1001
DROSHA	auto	ok	36	50	14.29	1	no	0.1722
IGF1R	auto	ok	36	33.33	25	1	no	0.09049
FLT4	auto	ok	35	33.33	42.86	1	no	0.07874
TCF3	auto	ok	35	33.33	60	1	no	0.1127
RICTOR	auto	ok	34	60	55.56	1	no	0.1368
KDM5C	auto	ok	33	100	66.67	1	no	0.08492
PTPRT	auto	ok	33	100	60	1	no	0.08338

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
RUNX1	auto	ok	33	50	33.33	1	no	0.192
BCL2L11	auto	ok	32	23.08	21.43	1	no	0.2968
DNMT3A	auto	ok	32	100	36.36	1	no	0.2238
PIK3R2	auto	ok	32	20	40	1	no	0.1289
RPS6KB2	auto	ok	32	50	0	1	no	0.109
ERCC2	auto	ok	31	50	75	1	no	0.1011
INSR	auto	ok	31	50	50	1	no	0.08249
PDGFRA	auto	ok	31	83.33	55.56	1	no	0.1285
TP53BP1	auto	ok	31	66.67	66.67	1	no	0.1592
XPO1	auto	ok	31	12.5	41.67	1	no	0.2475
ESR1	auto	ok	30	37.5	38.46	1	no	0.1397
ETV1	auto	ok	30	68.75	0	1	no	0.3961
GNAS	auto	ok	30	40	0	1	no	0.1616
AXIN1	auto	ok	29	100	66.67	1	no	0.2859
BICC1	auto	ok	29	100	67.86	1	no	0.3992
CALR	auto	ok	29	80	0	1	no	0.1558
CBL	auto	ok	29	50	50	1	no	0.2599
FH	auto	ok	29	16.67	13.33	1	no	0.2407
FUBP1	auto	ok	29	50	66.67	1	no	0.09592
JAK1	auto	ok	29	50	0	1	no	0.1105
KDR	auto	ok	28	25	33.33	1	no	0.2071
RARA	auto	ok	28	66.67	35.71	1	no	0.07262
SLX4	auto	ok	28	71.43	61.11	1	no	0.3416
TET1	auto	ok	28	100	28.57	1	no	0.07472
IKBKE	auto	ok	27	50	36.36	1	no	0.1335
MAP2K4	auto	ok	27	50	0	1	no	0.2269
PPM1D	auto	ok	27	25	25	1	no	0.2456
SMAD3	auto	ok	27	33.33	55.56	1	no	0.1027
STAT5B	auto	ok	27	66.67	75	1	no	0.1365
TRAF7	auto	ok	27	25	20	1	no	0.1797
IRS2	auto	ok	26	50	77.78	1	no	0.08323
UPF1	auto	ok	26	100	60	1	no	0.08901
FLT1	auto	ok	25	33.33	33.33	1	no	0.1524
MALT1	auto	ok	25	100	42.86	1	no	0.2043
PAK1	auto	ok	25	16.67	10	1	no	0.1255
ASXL2	auto	ok	24	66.67	57.14	1	no	0.1281
DIS3	auto	ok	24	50	16.67	1	no	0.1987

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
TP63	auto	ok	24	66.67	14.29	1	no	0.1073
MDM2	auto	ok	23	66.67	47.06	1	no	0.1142
RAF1	auto	ok	23	37.5	50	1	no	0.0757
STAT5A	auto	ok	23	33.33	0	1	no	0.05923
AKT2	auto	ok	22	25	22.22	1	no	0.06328
PARP1	auto	ok	22	20	40	1	no	0.1518
PIK3CD	auto	ok	22	33.33	50	1	no	0.09106
PLK2	auto	ok	22	50	57.14	1	no	0.3243
PPP2R1A	auto	ok	22	75	33.33	1	no	0.1207
PRKCI	auto	ok	22	66.67	50	1	no	0.1474
MAP2K2	auto	ok	21	40	14.29	1	no	0.1855
RPS6KA4	auto	ok	21	62.5	58.33	1	no	0.102
SMARCD1	auto	ok	21	33.33	57.14	1	no	0.1512
ASPSCR1	auto	ok	20	93.33	20	1	no	0.4824
ERF	auto	ok	20	50	28.57	1	no	0.1636
FLCN	auto	ok	20	71.43	44.44	1	no	0.1988
PIK3R3	auto	ok	20	100	0	1	no	0.1712
SMYD3	auto	ok	20	50	33.33	1	no	0.1125
SRC	auto	ok	20	100	0	1	no	0.1138
CASP8	auto	ok	19	66.67	55.56	1	no	0.1126
CHEK2	auto	ok	19	25	16.67	1	no	0.2676
LATS1	auto	ok	19	50	33.33	1	no	0.1419
YAP1	auto	ok	19	50	20	1	no	0.1214
BCAS3	auto	ok	18	16.67	46.67	1	no	0.1619
FGFR4	auto	ok	18	50	44.44	1	no	0.1469
PGR	auto	ok	18	33.33	57.14	1	no	0.09687
VTCN1	auto	ok	18	50	60	1	no	0.1502
CSF3R	auto	ok	17	66.67	66.67	1	no	0.1373
EGFL7	auto	ok	17	66.67	25	1	no	0.2599
LDLR	auto	ok	17	37.5	61.54	1	no	0.3481
LMNA	auto	ok	17	100	38.46	1	no	0.3977
PLCG2	auto	ok	17	40	66.67	1	no	0.1599
SDC4	auto	ok	17	100	0	1	no	0.5756
SEC16A	auto	ok	17	80	50	1	no	0.4034
SND1	auto	ok	17	100	17.65	1	no	0.4507
SUFU	auto	ok	17	66.67	25	1	no	0.1131
TGFBR1	auto	ok	17	100	75	1	no	0.2072

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
CUL1	auto	ok	16	100	0	1	no	0.3752
NRG1	auto	ok	16	100	86.67	1	no	0.2928
PDGFRB	auto	ok	16	100	40	1	no	0.1815
PRKD1	auto	ok	16	66.67	25	1	no	0.1397
TRAP1	auto	ok	16	80	37.5	1	no	0.3595
CCNE1	auto	ok	15	50	0	1	no	0.1214
CREB3L1	auto	ok	15	100	100	1	no	0.5383
TNFRSF14	auto	ok	15	100	0	1	no	0.2625
TTC28	auto	ok	15	28.57	63.64	1	no	0.1601
AGK	auto	ok	14	100	14.29	1	no	0.543
CDK4	auto	ok	14	66.67	0	1	no	0.1183
FOXO1	auto	ok	14	100	0	1	no	0.2625
NUF2	auto	ok	14	100	11.11	1	no	0.2164
SESN3	auto	ok	14	50	0	1	no	0.1214
SF3B1	auto	ok	14	33.33	40	1	no	0.1227
TRAF2	auto	ok	14	33.33	20	1	no	0.1634
EZR	auto	ok	13	66.67	0	1	no	0.4088
MST1R	auto	ok	13	33.33	40	1	no	0.1109
NPM1	auto	ok	13	100	50	1	no	0.1521
TRAPPC9	auto	ok	13	42.86	0	1	no	0.4838
YES1	auto	ok	13	50	16.67	1	no	0.2015
AKT3	auto	ok	12	100	50	1	no	0.2639
FYN	auto	ok	12	100	66.67	1	no	0.1446
IDH1	auto	ok	12	100	0	1	no	0.2625
PAX5	auto	ok	12	100	66.67	1	no	0.1874
PIK3C3	auto	ok	12	100	100	1	no	0.3862
PRKAR1A	auto	ok	12	66.67	20	1	no	0.08571
SDHA	auto	ok	12	66.67	60	1	no	0.1572
ERCC4	auto	ok	11	33.33	50	1	no	0.1227
GSK3B	auto	ok	11	100	0	1	no	0.2625
MKRN1	auto	ok	11	77.78	0	1	no	0.3812
MLH1	auto	ok	11	100	50	1	no	0.2625
PHF7	auto	ok	11	100	0	1	no	0.2625
RAD51D	auto	ok	11	100	0	1	no	0.2625
SOS1	auto	ok	11	50	0	1	no	0.1471
TACC2	auto	ok	11	100	100	1	no	0.5439
BMPR1A	auto	ok	10	100	0	1	no	0.3792

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
CDK8	auto	ok	10	100	0	1	no	0.1472
CUL3	auto	ok	10	50	0	1	no	0.2599
IKZF3	auto	ok	10	50	62.5	1	no	0.3442
MDM4	auto	ok	10	50	50	1	no	0.1524
RXRA	auto	ok	10	50	66.67	1	no	0.1455
TRIM24	auto	ok	10	100	0	1	no	0.3899
ARAF	auto	ok	9	100	50	1	no	0.2139
NTHL1	auto	ok	9	50	66.67	1	no	0.1988
PPARG	auto	ok	9	100	50	1	no	0.2036
RAD51	auto	ok	9	100	66.67	1	no	0.2788
STK40	auto	ok	9	33.33	0	1	no	0.1953
ACOXL	auto	ok	8	14.29	50	1	no	0.3113
ETV4	auto	ok	8	100	16.67	1	no	0.2834
FBXL20	auto	ok	8	100	28.57	1	no	0.4366
IKZF1	auto	ok	8	100	0	1	no	0.3949
KRAS	auto	ok	8	100	0	1	no	0.1856
MAPK1	auto	ok	8	50	0	1	no	0.2599
MSI1	auto	ok	8	25	40	1	no	0.1748
MTAP	auto	ok	8	33.33	25	1	no	0.4254
MYD88	auto	ok	8	66.67	33.33	1	no	0.1475
PDE4D	auto	ok	8	25	66.67	1	no	0.3443
PTK2	auto	ok	8	66.67	50	1	no	0.2622
STRN	auto	ok	8	100	0	1	no	0.4356
CACNA1A	auto	ok	7	33.33	50	1	no	0.07545
CDK6	auto	ok	7	50	0	1	no	0.2599
CREB1	auto	ok	7	80	71.43	1	no	0.4162
DCUN1D1	auto	ok	7	100	0	1	no	0.1611
DNM2	auto	ok	7	40	42.86	1	no	0.1853
GAB2	auto	ok	7	50	20	1	no	0.1214
IGF2	auto	ok	7	100	75	1	no	0.2013
KREMEN1	auto	ok	7	40	42.86	1	no	0.2054
NR4A3	auto	ok	7	100	71.43	1	no	0.4192
RAB11FIP4	auto	ok	7	50	28.57	1	no	0.4279
ST7	auto	ok	7	33.33	0	1	no	0.437
TACC1	auto	ok	7	100	100	1	no	0.268
USP34	auto	ok	7	33.33	71.43	1	no	0.5038
CAMTA1	auto	ok	6	100	0	1	no	0.1241

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
CDH3	auto	ok	6	33.33	25	1	no	0.2917
CDKN2C	auto	ok	6	100	66.67	1	no	0.1529
CPM	auto	ok	6	50	0	1	no	0.1385
CTNNA3	auto	ok	6	75	0	1	no	0.2593
DTNB	auto	ok	6	100	0	1	no	0.4164
EIF3H	auto	ok	6	100	0	1	no	0.529
EPHX3	auto	ok	6	100	0	1	no	0.2625
GOPC	auto	ok	6	75	50	1	no	0.5455
SMARCA2	auto	ok	6	50	66.67	1	no	0.147
TANC2	auto	ok	6	100	50	1	no	0.1423
TFG	auto	ok	6	100	100	1	no	0.3411
ZNRF3	auto	ok	6	50	66.67	1	no	0.3451
ATE1	auto	ok	5	50	0	1	no	0.3752
BCL2L1	auto	ok	5	50	50	1	no	0.2008
CNTNAP2	auto	ok	5	66.67	0	1	no	0.2602
CREM	auto	ok	5	20	100	1	no	0.252
EIF4E	auto	ok	5	50	0	1	no	0.2599
EMID1	auto	ok	5	75	25	1	no	0.5006
FBXL7	auto	ok	5	100	75	1	no	0.2627
GPATCH8	auto	ok	5	50	50	1	no	0.1669
HCN1	auto	ok	5	100	0	1	no	0.3772
LRP1B	auto	ok	5	100	100	1	no	0.2062
MAD1L1	auto	ok	5	40	0	1	no	0.226
NELL1	auto	ok	5	50	80	1	no	0.2507
PRKAG2	auto	ok	5	25	50	1	no	0.3136
RAC2	auto	ok	5	50	0	1	no	0.2599
SDHAF2	auto	ok	5	100	0	1	no	0.4933
SFPQ	auto	ok	5	50	75	1	no	0.4933
SMARCE1	auto	ok	5	100	0	1	no	0.4933
SMURF2	auto	ok	5	100	75	1	no	0.202
SPTBN1	auto	ok	5	100	0	1	no	0.3932
THSD4	auto	ok	5	50	25	1	no	0.1105
TPR	auto	ok	5	100	0	1	no	0.5142
CDKN2A	auto	ok	185	14.29	0			0.2857
RB1	auto	ok	147	0	69.23			0.2149
APC	auto	ok	110	0	45.45			0.3244
FAT1	auto	ok	95	0	30			0.188

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
TERT	auto	ok	92	0	81.25			0.3993
ZFHX3	auto	ok	85	0	16.67			0.2857
NCOR1	auto	ok	80	0	35.29			0.3513
ARID1B	auto	ok	72	0	58.33			0.2824
ARID2	auto	ok	72	0	23.08			0.2429
CIC	auto	ok	68	0	47.37			0.1588
SMAD4	auto	ok	62	0	16.67			0.2106
SPEN	auto	ok	61	0	27.78			0.1579
NSD3	auto	ok	58	0	0			0.1429
ANKRD11	auto	ok	54	0	66.67			0.3333
FOXA1	auto	ok	51	0	20			0.4607
ATR	auto	ok	48	0	33.33			0.1588
BRIP1	auto	ok	46	0	23.53			0.2095
SETD2	auto	ok	46	0	50			0.2621
NOTCH4	auto	ok	44	0	68.75			0.1325
B2M	auto	ok	42	0	0			0.6481
DNMT3B	auto	ok	40	0	16.67			0.2659
KEAP1	auto	ok	40	0	33.33			0.1454
NFE2L2	auto	ok	40	0	20			0.3516
PRKN	auto	ok	40	0	0			0.25
NBN	auto	ok	38	0	33.33			0.2346
NSD2	auto	ok	37	0	0			0.3333
PREX2	auto	ok	37	0	50			0.2391
BCOR	auto	ok	36	0	57.14			0.1588
MAP3K14	unresolved	failed	36					
TBX3	auto	ok	36	0	60			0.1766
ASXL1	auto	ok	35	0	100			0.2222
JAK3	auto	ok	32	0	37.5			0.164
RASA1	auto	ok	32	0	66.67			0.5
ARID5B	auto	ok	31	0	33.33			0.3356
CSDE1	auto	ok	31	0	60			0.4
STAG2	auto	ok	31	0	0			0.5
TOP1	auto	ok	31	0	50			0.2306
AR	auto	ok	30	0	50			0.5
EZH1	auto	ok	30	0	20			0.205
NTRK2	auto	ok	29	0	40			0.2306
AXIN2	auto	ok	28	0	75			0.4266

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
CDC73	auto	ok	28	0	50			0.5
TEK	auto	ok	28	0	55.56			0.135
TSC1	auto	ok	28	0	57.14			0.4311
JAK2	auto	ok	27	0	33.33			0.3016
SOX9	auto	ok	26	0	25			0.2356
CARD11	auto	ok	25	0	28.57			0.2168
CDKN1B	auto	ok	24	0	0			0.3016
AKT1	auto	ok	23	0	42.86			0.2857
MAP3K13	auto	ok	23	0	20			0.3686
MED12	auto	ok	23	0	33.33			0.3016
MSH2	auto	ok	23	0	0			0.2731
NFKBIA	auto	ok	23	0	100			1
SMARCB1	auto	ok	23	0	60			0.4
TCF7L2	auto	ok	23	0	50			0.25
EIF4A2	auto	ok	22	0	33.33			0.4266
MEN1	auto	ok	22	0	0			0.3016
MYC	auto	ok	22	0	0			0.3333
RAD54L	auto	ok	22	0	60			0.2016
ABL1	auto	ok	21	0	25			0.2183
FLT3	auto	ok	21	0	20			0.2356
TET2	auto	ok	21	0	66.67			0.5516
BLM	auto	ok	20	0	50			0.1936
KIT	auto	ok	20	0	0			0.4266
MSH6	auto	ok	20	0	80			0.3016
PIK3CB	auto	ok	20	0	60			0.3516
CDKN2B-AS1	non_coding	failed	19					
CSF1R	auto	ok	19	0	33.33			0.135
EPHA3	auto	ok	19	0	0			0.4266
GRIN2A	auto	ok	19	0	20			0.5516
NEGR1	auto	ok	19	0	66.67			0.3333
PALB2	auto	ok	19	0	0			0.2
PTCH1	auto	ok	19	0	20			0.2186
SEPTIN14	auto	ok	19	0	0			1
BBC3	auto	ok	18	100	0			0.375
GATA3	auto	ok	18	0	0			0.5
AMER1	auto	ok	17	0	50			0.3333
COP1	auto	ok	17	0	0			0.5

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
DDR2	auto	ok	17	0	0			0.2016
EPHA5	auto	ok	17	0	33.33			0.5516
ERCC5	auto	ok	17	0	60			0.375
INPP4B	auto	ok	17	100	0			1
RAD50	auto	ok	17	0	33.33			0.3016
TAP1	auto	ok	17	0	50			0.1429
BCL6	auto	ok	16	0	66.67			0.3016
CTCF	auto	ok	16	0	50			0.25
DICER1	auto	ok	16	0	0			0.3856
ERRFI1	auto	ok	16	0	0			0.6667
LATS2	auto	ok	16	0	0			0.8016
MITF	auto	ok	16	0	100			1
MSI2	auto	ok	16	0	50			0.4266
NCOA4	auto	ok	16	0	0			1
RAD52	auto	ok	16	0	100			0.8016
RECQL	auto	ok	16	0	0			0.2731
SMO	auto	ok	16	0	12.5			0.3329
SUZ12	auto	ok	16	0	20			0.2016
TGFBR2	auto	ok	16	0	100			0.5
CCND1	auto	ok	15	0	57.14			0.1429
INPP4A	auto	ok	15	33.33	0			0.2857
MSH3	auto	ok	15	0	50			0.5
SRP19	auto	ok	15	0	0			1
E2F3	auto	ok	14	0	20			0.2016
GPS2	auto	ok	14	0	0			0.5
HGF	auto	ok	14	0	66.67			0.4266
LINC00114	non_coding	failed	14					
MPL	auto	ok	14	0	83.33			0.3775
NUP93	auto	ok	14	0	25			0.3516
RIT1	auto	ok	14	0	0			0.3596
TAP2	auto	ok	14	0	0			0.5
CD79A	auto	ok	13	0	37.5			0.25
CYLD	auto	ok	13	0	0			1
DAXX	auto	ok	13	0	0			1
MAP2K1	auto	ok	13	0	66.67			0.25
MEF2B	auto	ok	13	0	60			0.4
MRE11	auto	ok	13	0	100			0.2

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
PAK5	auto	ok	13	0	0			0.5
RAD51C	auto	ok	13	0	0			0.1766
SPOP	auto	ok	13	0	50			0.4266
BIRC3	auto	ok	12	0	66.67			0.3016
BTK	auto	ok	12	0	0			1
CBFB	auto	ok	12	0	0			1
CEBPA	auto	ok	12	0	0			0.1667
DUSP4	auto	ok	12	0	33.33			0.3016
EED	auto	ok	12	0	0			0.4266
IRS1	auto	ok	12	0	0			0.5
PIK3CA	auto	ok	12	0	80			0.3516
RAB35	auto	ok	12	0	0			0.3356
SMAD2	auto	ok	12	0	0			0.8016
VEGFA	auto	ok	12	0	0			0.4266
ALOX12B	auto	ok	11	0	0			0.4266
BABAM1	auto	ok	11	0	0			0.5
BCL2	auto	ok	11	0	0			1
EPHA7	auto	ok	11	0	50			0.4266
FANCC	auto	ok	11	0	0			0.5856
FBXW7	auto	ok	11	0	0			0.3016
FGF19	auto	ok	11	0	0			0.2731
IDH2	auto	ok	11	0	0			0.4266
SRSF2	auto	ok	11	0	100			1
TNFAIP3	auto	ok	11	0	0			0.3016
AURKA	auto	ok	10	0	33.33			0.5516
CCND2	auto	ok	10	0	100			0.3516
EPAS1	auto	ok	10	0	20			0.3516
EPHB1	auto	ok	10	0	100			0.5
H3C2	unresolved	failed	10					
MAPK3	auto	ok	10	0	0			0.8016
PIK3CG	auto	ok	10	0	0			
PMS1	auto	ok	10	0	0			
PTPN11	auto	ok	10	0	0			0.3333
SH2B3	auto	ok	10	0	33.33			0.25
SOX17	auto	ok	10	0	0			0.5516
SYK	auto	ok	10	0	0			1
CHEK1	auto	ok	9	0	0			0.4266

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
GATA1	auto	ok	9	0	0			1
IFNGR1	auto	ok	9	0	33.33			0.3016
LRP1	auto	ok	9	0	20			0.5016
MAPKAP1	auto	ok	9	0	100			0.8016
PPP6C	auto	ok	9	0	0			1
PRDM14	auto	ok	9	0	0			
REL	auto	ok	9	0	0			
SESN1	auto	ok	9	0	0			0.2016
SHQ1	auto	ok	9	0	100			0.5
AKAP8	auto	ok	8	0	0			1
AURKB	auto	ok	8	0	33.33			0.3333
BARD1	auto	ok	8	0	0			
BCL2L14	auto	ok	8	66.67	0			0.75
CCND3	auto	ok	8	0	0			0.8016
CD79B	auto	ok	8	0	33.33			0.3016
ERCC3	auto	ok	8	0	33.33			0.3016
FGF3	auto	ok	8	0	0			0.3333
GNA11	auto	ok	8	0	0			1
HNF1A	auto	ok	8	0	25			0.3516
ICOSLG	auto	ok	8	0	0			
IRF4	auto	ok	8	0	25			0.2
LYN	auto	ok	8	0	0			0.2391
MCL1	auto	ok	8	0	60			0.3333
NBPF20	auto	ok	8	0	25			0.3516
PDCD1	auto	ok	8	0	0			1
RRAS	auto	ok	8	0	33.33			0.3333
SLC25A21	auto	ok	8	0	14.29			0.6141
XRCC2	auto	ok	8	0	0			1
ACP3	auto	ok	7	0	0			0.4286
AP1B1	auto	ok	7	0	0			1
ASIC2	auto	ok	7	0	0			0.3333
ATP1B2	auto	ok	7	0	0			1
CRKL	auto	ok	7	0	50			0.5
DCBLD1	auto	ok	7	0	0			
DGKH	auto	ok	7	0	28.57			0.5874
EPCAM	auto	ok	7	0	0			0.5
H3C4	auto	ok	7	0	0			

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
INHA	auto	ok	7	0	0			1
JUN	auto	ok	7	0	0			
KLF4	auto	ok	7	0	0			
NKX3-1	auto	ok	7	0	0			1
PDPK1	auto	ok	7	0	100			1
RYBP	auto	ok	7	0	0			
SDHC	auto	ok	7	0	66.67			0.3016
SHOC2	auto	ok	7	0	100			1
TENT5C	auto	ok	7	0	0			
CD274	auto	ok	6	0	0			1
CD276	auto	ok	6	0	0			
FGF4	auto	ok	6	0	50			0.5
GALNT11	auto	ok	6	0	66.67			1
GRB7	auto	ok	6	0	33.33			0.8
H1-2	auto	ok	6	0	0			1
H3-3B	auto	ok	6	0	0			0.5
H3C6	unresolved	failed	6					
HOXB13	auto	ok	6	0	0			0.5
IL7R	auto	ok	6	0	100			0.5
INHBA	auto	ok	6	0	50			0.4266
ITM2B	auto	ok	6	0	50			0.8016
KANSL1	auto	ok	6	0	0			0.4
KAZN	auto	ok	6	0	66.67			0.2391
KLF5	auto	ok	6	0	0			
MAX	auto	ok	6	0	0			
NKX2-1	auto	ok	6	0	100			1

Showing the first 500 of 544 scanned genes.

Honorable mentions: highly ranked non-FDR-significant genes worth human review

Did NOT survive genome-wide FDR correction -- ranked by raw Fisher p-value among non-significant genes. Not a claim of significance.

Rank	Gene	Fisher p	Best q	N events	In-frame%	Domain-ret.%
1	CDKN2B	0.02222	0.5249	12	66.67	20
2	NTRK1	0.05145	0.3201	78	72.73	84.21

Rank	Gene	Fisher p	Best q	N events	In-frame%	Domain-ret.%
3	NTRK3	0.05556	0.07592	58	98.21	96.49
4	IKBKE	0.06667	1	12	50	36.36
5	FLI1	0.07688	1	118	68.22	94.02
6	RTEL1	0.1167	1	24	50	37.5
7	KDM5A	0.119	1	10	50	55.56
8	FH	0.1429	1	16	16.67	13.33
9	CDK12	0.1734	1	43	50	39.02
10	POLD1	0.1923	0.7353	16	33.33	46.15
11	AGO2	0.2252	1	23	46.15	31.82
12	NRG1	0.2273	1	15	100	86.67
13	MTAP	0.25	1	4	33.33	25
14	RPS6KA4	0.2619	1	13	62.5	58.33
15	PLK2	0.2857	1	7	50	57.14

Figures

manhattan

